What role could decision models play in GeoAI?

Qualifying Spatial Relations using Jev.

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Introduction

Janowicz et al., 2020 asked if there will be an *"artificial GIS analyst"* by 2030. Since then there have been huge strides both in the field of artificial intelligence (AI) and autonomous geoinformation systems (GIS) (Li et al., 2025). As spatial analysis requires large amounts of up-to-date geodata, GeoAI (Janowicz et al., 2020) requires substantial compute, leading to low accessibility and high environmental impact (Janowicz, 2023; Shi et al., 2025). Also, large language models (LLMs) may produce a high overhead in reasoning tokens (Chen et al., 2025; Wang et al., 2026) and a long-form response in natural language, which might be unnecessary and difficult to process downstream. *Decision models*, also described as *system-one* (Kahneman, 2011) models, might be more suitable for integration in GIS-tools, as they operate more efficiently and quicker than LLMs, as well as responding in a predetermined, consistent format. To test this assumption this thesis reproduces the experiment on *spatial relation qualification* done by Ji et al., 2025 using Typesafe AI's model Jev (Almeida, 2026) to enable a direct comparison of the two models.

Research Goals

The aim of this thesis is to do an initial exploration of the model Jev, in the hopes of gaining broader insights of decision models in spatial settings. As Jev is not a chat-based model, proven prompt engineering methods, like few-shot learning or chain-of-thought reasoning (Schulhoff et al., 2025), might not have the same advantages as with LLMs. This will be tested using the same prompting strategies as in the original experiment. Ethical points, especially sustainability, but also briefly bias (Aldahoul and Zaki, 2026), will be discussed as well. In total the thesis links up with works about GeoAI (Janowicz et al., 2020; Ji et al., 2025), autonomous GIS (Li et al., 2025), and ethical use of GeoAI (Janowicz, 2023; Shi et al., 2025).

The following research questions will be addressed:

- How does the decision model Jev compare with ChatGPT-4 in the task of spatial relation qualification in terms of accuracy and cost?
- In what ways could decision models benefit GeoAI, especially considering aspects of sustainability?
- Do proven prompt engineering methods like few-shot prompting and chain-of-thought reasoning work for system-one intelligence models?

Methodology

To ensure a good comparison the thesis will stay as true as possible to the original experiment in Ji et al., 2025, using the same data and setup. The prompts will have to be adapted to the unique requirements of Jev, but the same strategies (few-shot prompting and chain-of-thought reasoning) will be tested. The experiment will utilize Typesafe AI's API and Python SDK to access Jev 1.13.0. All code and results will be made available together with the thesis on Github. A short review of related works will accompany the experiment and discussion.

Expected Results and Challenges

Initial experimentation proved Jev's speed and (cost-)efficiency, while it could not beat the benchmark set by ChatGPT-4 in terms of accuracy. Still, the confidence score, which Jev assigns to its answers, seems reliable, and some interesting patterns occurred in the initial results, which merit further investigation.

The main challenge this thesis will face is a problem common with all LLMs: Their opacity. Jev's answers lack any reasoning and cannot be reproduced reliably. In addition, Jev is a proprietary model, with very little official information released publicly.

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